■ MedAssist AI Platform

Comprehensive User Operation Guide, Features, System Architecture & ML Methodology

System Version: 2.0 Enterprise	**Release Date:** July 2026
Core Framework: Next.js 16 + FastAPI ML Backend	**ML Model:** 200-Tree Random Forest (658 Classes)

1. How to Use the Website & User Interaction Guide

MedAssist AI provides an intuitive, patient-centered diagnostic interface designed for simple navigation and fast clinical insights. Users interact with the platform through three primary modules:

Step 1: Accessing the Platform

Navigate to http://localhost:6700 in any standard web browser to enter the MedAssist AI portal.

Step 2: AI Assistant Chat & Persistent Brain Memory

- Open **/ai-assistant** from the navigation bar.
- Select a clinical prompt pill (e.g. *Analyze Symptoms*, *Drug Interactions*, *Lab Report Summary*).
- The AI Assistant remembers past patient conversations across sessions using its **Neural Brain Memory Engine**.
- Type *'generate PDF'* at any time during chat to receive an interactive, downloadable clinical PDF report.

Step 3: AI Symptom Checker & Mode Options

- Open **/dashboard/symptom-checker** or **/dashboard/analysis**.
- Use the top **Mode Switcher Bar** to toggle between:
  - **Option 1 (All-in-One Checker):** Single-page form with Patient Details, Symptom Selector, Duration/Severity slider, and Upload Reports button.
  - **Option 2 (Step-by-Step Wizard):** 4-step guided assessment (Patient Info → Vitals → Lifestyle → Symptoms).
- Click **'Upload Reports / Scans'** to attach PDF lab reports or image medical scans for automatic text extraction.
- Click **'Analyze Symptoms'** to query the 200-Tree Random Forest Machine Learning backend.

2. Core Features, Services & Patient Benefits

Feature / Service	Description & Capability	Patient Benefit
200-Tree ML Model	RandomForest Classifier trained on 60,000 samples across 658 disease classes & 377 symptom features.	Accurate multi-class differential diagnosis with Top-1, Top-3, & Top-5 rankings.
Neural Brain Memory	Cross-session memory scanner aggregating historical chat logs from server JSON database.	Continuous care context without repeating medical history.
Dual Mode Switcher	Seamless toggle between Option 1 (All-in-One) and Option 2 (Step-by-Step Wizard).	Flexible user experience tailored to user preference.

Report Parser & OCR	PDF and Image parser extracting clinical text and pre-selecting symptom pills automatically.	Zero manual typing required for lab reports.
Rich Media PDF Engine	React Portal modal generating styled PDF reports with autoTables, flowcharts, and links.	Instant printable summary for doctor consultations.

3. Frameworks, Tools & Technology Stack

Layer	Technologies Used
Frontend Core	Next.js 16 (App Router, Turbopack), React 19, TypeScript
UI & Styling	TailwindCSS, Glassmorphism, Framer Motion, Lucide React Icons
ML Backend	Python FastAPI, Scikit-learn, Joblib, Pandas, NumPy, Uvicorn (Port 8000)
LLM & AI Engine	Groq API (llama-3.3-70b-versatile) with Auto-Recovery Model Cascade
PDF & Media	jsPDF, jspdf-autotable, ReportLab, PDF2JSON, Canvas Vision API
Data Storage	Server-Side Public JSON Store (public/data/), Firebase Firestore

4. Machine Learning Methodology & Performance Results

The core diagnostic prediction engine is powered by a **200-Tree Random Forest Classifier** trained on a dataset of **60,000 patient records** across **377 symptom features** and **658 disease categories**.

Metric Name	Weighted Score	Macro Score	Description
Overall Accuracy	**74.72%**	**74.72%**	Exact primary disease match
Precision	**78.99%**	**64.97%**	Low false positive rate
Recall	**74.72%**	**48.93%**	True positive sensitivity
F1 Score	**74.11%**	**53.01%**	Harmonic mean of precision & recall
■ Top-1 Accuracy	**74.72%**	-	Primary predicted condition
■ Top-3 Accuracy	**86.72%**	-	Disease present in top 3 ranks
■ Top-5 Accuracy	**89.83%**	-	Disease present in top 5 ranks

5. Complete Project Folder Architecture

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D:\infosys_project\
■■■ frontend/ (Next.js 16 Web Application)
■ ■■■ src/app/ (App Router Pages & Next.js API Routes)
■ ■ ■■■ (public)/ai-assistant/page.tsx (AI Assistant & Chat)
■ ■ ■■■ (dashboard)/dashboard/symptom-checker/page.tsx (Dual Mode Option 1 & 2)
■ ■ ■■■ (dashboard)/dashboard/analysis/page.tsx (Step Wizard Page)
■ ■ ■■■ api/predict/route.ts (Connects to Python ML Backend)
■ ■■■ src/components/chat/ (PDF Document Card, Sidebar, Suggestions)
■ ■■■ src/components/analysis/ (Wizard Steps & Result Dashboard)
■ ■■■ src/lib/brain-memory.ts (Cross-Session Memory Scanner)
■ ■■■ src/lib/pdf-generator.ts (Rich Media PDF Generator)
■ ■■■ public/data/ (conversations.json & messages.json server DB)
■■■ ml_backend/ (Python FastAPI Microservice)
■ ■■■ app.py (FastAPI predict endpoint on Port 8000)
■ ■■■ best_model.joblib (Trained 200-Tree Random Forest)
■ ■■■ label_encoder.joblib (Pickled LabelEncoder)
■ ■■■ feature_names.json (377 Feature Schema)
■■■ backend/ (Legacy Express API & Auth Services)
```

6. Diagnostic Flowcharts & Interaction Pathways

[USER INPUT / REPORT UPLOAD]
↓
[PDF/IMAGE PARSER OCR] → *Auto-selects Symptom Pills*
↓
[OPTION SWITCHER: All-in-One vs Step Wizard]
↓
[NEXT.JS API ROUTE: /api/predict]
↓
[PYTHON FASTAPI ML BACKEND (Port 8000)]
↓
[200-TREE RANDOM FOREST MODEL (658 Disease Classes)]
↓
[TOP-5 DIFFERENTIAL DIAGNOSES + RISK LEVEL + URGENCY]
↓
[INTERACTIVE REACT PORTAL PDF EXPORTER]